



Complete Technical Documentation

AI-Driven Hyper-Local Business Advisory and Financial Structuring Assistant
for Rural Micro-Entrepreneurs

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Part 1 — Big Picture

What BizNex Does

BizNex is an AI-powered business advisory platform for rural Indian micro-entrepreneurs. It helps them evaluate market feasibility, find government loan schemes, calculate funding eligibility, and get personalized business advice — all in multiple Indian languages.

User Journey (Story)

User opens BizNex → Sees Landing Page with 3-step workflow (Understand Market → Structure Funding → Build With Confidence) → Can select language from dropdown (English, Hindi, Telugu, Tamil, Kannada, Marathi) → Clicks "Get Started" → Register Page → Registers with: Name, Email, Password, Mobile, Village, District, State, Language → Supabase creates auth account + profile row in database → Redirects to Dashboard **Dashboard shows:** Welcome message, business profile count, quick insight cards **User clicks "Create Business Profile"** → 5-step wizard: 1. Location – India-wide search via Nominatim (OpenStreetMap) with GPS support 2. Business Type – category selection 3. Investment – preset amounts or custom entry 4. New or Existing Business – toggle 5. Review & Create – summary with detected coordinates → Profile saved to localStorage (includes structured location data: village, block, district, state, lat/lng) **User can now access all features:** — Market Analysis → AI analyzes demand, competition, revenue — Business Plan → AI generates full plan — Scheme Finder → AI matches government schemes — Loan Calculator → AI calculates EMI & eligibility — Funding Advisor → AI recommends funding structure — Nearby Competitors → Real data from OpenStreetMap Overpass API + Leaflet map — Wholesale & Suppliers → Material requirements, sourcing plans, nearby suppliers via Overpass API — Local Insights → AI provides area demographics + Census data — AI Assistant → Chat with AI (context-aware deterministic responses when API unavailable) — Document Verification → Upload & verify docs (mock) — Admin Panel → (admin only) manage users, schemes, prompts

Data Flow

```
React Frontend (Browser)
  ↓ user input
Business Context (localStorage) + Auth Context (Supabase)
  ↓ API calls
src/lib/ai.ts → callServerAI() → /api/ai (Vercel serverless) → Groq API (primary) → Gemini API (fallback) →
Deterministic engine (last resort)
  ↓ external data
Nominatim (OpenStreetMap) → Location search & geocoding (free, no API key)
Overpass API (OpenStreetMap) → Nearby competitors & suppliers (free, no API key)
India Census Data → District-level demographics (built-in)
  ↓ AI response
React Components render charts, cards, tables
```

Part 2 — Tech Stack

Technology	What It Does	Where In Project
React 18	UI library — builds all pages as components	Every file in src/
TypeScript	Type-safe JavaScript — catches errors at build time	Every .tsx/.ts file
Vite	Dev server + build tool — fast HMR, bundling	vite.config.ts, package.json
Tailwind CSS	Utility-first CSS — all styling via class names	tailwind.config.js, every component
React Router v6	Client-side routing — page navigation	src/App.tsx (Routes, Route)

Supabase JS	Auth + PostgreSQL database	src/lib/supabase.ts, AuthContext.tsx
Groq API	Primary AI — ultra-fast LLM inference (Qwen 3.8-27B), server-side only	api/ai.ts (Vercel serverless)
Google Gemini API	Fallback AI — Gemini 3.6 Flash, server-side only	api/ai.ts (Vercel serverless)
Axios	HTTP client — makes API calls to AI providers and from frontend to /api	src/lib/ai.ts, api/ai.ts
Vercel Serverless	API route — proxies AI calls, keeps API keys server-side	api/ai.ts (/api/ai endpoint)
Recharts	Charts — line, bar, pie, radar charts	MarketAnalysis, FundingAdvisor, LoanCalculator, etc.
Leaflet + React-Leaflet	Interactive maps — shows competitors on a map	NearbyCompetitors.tsx
Framer Motion	Animations — page transitions, reveals, hover effects	SchemeFinder, FundingAdvisor, etc.
Lucide React	Icons — all UI icons	Every component
react-hot-toast	Toast notifications — success/error messages	Login, Register, BusinessProfile
jsPDF	PDF generation — competitor report download	NearbyCompetitors.tsx
docx	Word document generation — business plan export	BusinessPlan.tsx
Nominatim (OpenStreetMap)	Free geocoding — location search, reverse geocoding	LocationSearch.tsx (no API key needed)
Overpass API (OpenStreetMap)	Nearby business discovery — competitors & suppliers	NearbyCompetitors.tsx, WholesaleFinder.tsx
react-i18next	Internationalization — 6 language translations	src/lib/i18n.ts, locales/*.json, DashboardLayout

What Happens If Each Is Removed

- **React** → Nothing renders
- **TypeScript** → Build errors
- **Tailwind** → All styling breaks
- **Supabase** → No authentication or database
- **Groq/Gemini** → AI features return deterministic responses from built-in financial engine
- **Recharts** → No charts on any feature page
- **Leaflet** → No competitor map
- **Framer Motion** → No animations (functional but static)

Part 3 — Project Folder Explanation

```
biznex/
├── api/
│   └── ai.ts          → Vercel serverless API route – proxies AI calls (Groq, Gemini)
├── src/
│   ├── main.tsx       → Entry point – mounts React app, wraps in providers
│   ├── App.tsx        → All routes (public + protected + admin)
│   ├── contexts/      → React Context providers (global state)
│   │   ├── AuthContext.tsx → User auth state (login, signup, demo, profile)
│   │   ├── BusinessContext.tsx → Business profiles (CRUD, active profile, seed data)
│   │   └── ThemeContext.tsx → Dark/light mode toggle
│   ├── lib/           → Utility/library files
│   │   ├── ai.ts      → Frontend AI calls → calls /api/ai serverless route
│   │   ├── supabase.ts → Supabase client initialization
│   │   ├── chartColors.ts → Theme-aware hex colors for recharts
│   │   ├── i18n.ts     → Internationalization setup (react-i18next)
│   │   ├── census-data.ts → District-level Census demographic data
│   │   ├── schemes-data.ts → 18+ structured government scheme database + eligibility scoring
│   │   ├── wholesale-data.ts → Material requirements + Overpass supplier discovery
│   │   ├── overpass.ts → Overpass API queries for real nearby business discovery
│   │   ├── geocoding.ts → Nominatim geocoding + reverse geocoding (free, no key)
│   │   └── locales/    → Translation files (en.json, hi.json, te.json, ta.json, kn.json, mr.json)
│   ├── components/
│   │   ├── ui/         → Reusable UI primitives
│   │   │   ├── Button.tsx → Styled button with variants
│   │   │   ├── Input.tsx  → Form input with icon support
│   │   │   ├── TextArea.tsx → Multi-line text input
│   │   │   ├── Select.tsx → Dropdown select
│   │   │   ├── Card.tsx   → Container card
│   │   │   ├── Modal.tsx  → Overlay modal dialog
│   │   │   └── LocationSearch.tsx → India-wide location autocomplete (Nominatim, GPS)
│   │   ├── layout/     → App shell components
│   │   │   ├── Layout.tsx → Public page shell (Navbar + Footer)
│   │   │   ├── DashboardLayout.tsx → Dashboard shell (Sidebar + Header + Main)
│   │   │   ├── Navbar.tsx → Top navigation bar
│   │   │   └── Footer.tsx → Page footer
│   │   ├── landing/    → Landing page sections
│   │   └── react-bits/ → Animation/effect components
│   │       ├── ScrollReveal.tsx, CountUp.tsx, GlowCard.tsx
│   │       ├── ParticlesBg.tsx, BlurText.tsx, TextType.tsx
│   │       ├── GradientText.tsx, GlassSurface.tsx
│   │       ├── MagneticButton.tsx, AnimatedList.tsx
│   │       └── index.ts → Re-exports all
├── pages/
│   ├── LandingPage.tsx → Homepage (hero, features, team, CTA)
│   ├── Dashboard.tsx   → Main dashboard (welcome, quick insights)
│   ├── BusinessProfile.tsx → CRUD business profiles (4-step wizard)
│   ├── Profile.tsx     → User profile page
│   ├── ComponentShowcase.tsx → UI component demo (dev only)
│   ├── auth/
│   │   ├── LoginPage.tsx → Email/password login + Demo Login button
│   │   └── RegisterPage.tsx → Full registration form
│   ├── features/
│   │   └── MarketAnalysis.tsx → AI market analysis with charts
```

```
├── BusinessPlan.tsx      → AI business plan generator
├── SchemeFinder.tsx      → AI government scheme matcher
├── LoanCalculator.tsx    → AI loan EMI calculator
├── FundingAdvisor.tsx    → AI funding structure advisor
├── NearbyCompetitors.tsx → Overpass API competitor finder + Leaflet map
├── WholesaleFinder.tsx   → Material requirements + Overpass supplier discovery
├── InsightsEngine.tsx    → AI local area insights + Census data
├── AIAssistant.tsx       → AI chat interface (context-aware deterministic fallbacks)
├── DocumentVerification.tsx → Doc upload + mock verification
├── admin/
│   └── AdminPanel.tsx → Admin dashboard (users, schemes, prompts)
├── supabase/
│   └── schema.sql       → Full database schema (tables, RLS, triggers)
├── public/              → Static assets (logos, favicon)
├── package.json          → Dependencies + scripts
├── vite.config.ts        → Vite build config
├── tailwind.config.js    → Tailwind customization (colors, fonts)
├── tsconfig.json         → TypeScript config
└── .env.example          → API key template
```

Part 4 — Components

Component	Purpose	Where Used	What It Displays
Button	Reusable button with variants	Every page	Clickable buttons
Input	Form input with label + icon	Forms (Login, Register, etc.)	Text/number inputs
TextArea	Multi-line input	BusinessPlan, AdminPanel	Text areas
Select	Dropdown select	BusinessProfile, SchemeFinder	Dropdown menus
Card	Container with border/shadow	All feature pages	Content containers
Modal	Overlay dialog	AdminPanel	Pop-up dialogs
Navbar	Top navigation	Public pages	Logo, nav links, theme toggle, language selector (6 languages)
Footer	Page footer	Public pages	Brand info, links
Layout	Public page wrapper	Landing, Login, Register	Navbar + content + Footer
DashboardLayout	Dashboard wrapper	All protected pages	Sidebar (with language switcher) + header (with business switcher) + content
ScrollReveal	Scroll-triggered fade-in	Landing page, Admin	Animated entry
GlowCard	Card with hover glow effect	AdminPanel, Landing	Highlighted stats
CountUp	Animated number counter	AdminPanel	Animated stat numbers
ParticlesBg	Background particle animation	Landing page	Visual effects
MessageBubble	Chat message with markdown	AIAssistant	AI/user chat bubbles
TypingIndicator	Bouncing dots	AIAssistant	"Thinking..." animation
LocationSearch	India-wide location autocomplete with GPS	BusinessProfile (onboarding)	Search results dropdown, current location button, coordinates display

Component Communication

- **Contexts** (`AuthContext` , `BusinessContext` , `ThemeContext`) provide global state via `useAuth()` , `useBusiness()` , `useTheme()` hooks
- **Pages** consume contexts + call AI functions from `src/lib/ai.ts`
- **UI components** are stateless, receive props only

Part 5 — Pages

Page	Route	Purpose	APIs Called
LandingPage	/	Marketing homepage	None
LoginPage	/login	Authentication	signIn() (Supabase) or demoLogin() (local)
RegisterPage	/register	New account creation	signUp() (Supabase)
Dashboard	/dashboard	Main hub	None (reads from context)
BusinessProfile	/business-profile	Manage business profiles (5-step wizard)	localStorage + Nominatim geocoding
Profile	/profile	User account info	updateProfile() (Supabase)
MarketAnalysis	/market-analysis	AI market analysis	analyzeMarket() → Groq/Gemini
BusinessPlan	/business-plan	AI business plan	generateBusinessPlan() → Groq/Gemini
SchemeFinder	/scheme-finder	Find gov schemes	findSchemes() → Groq/Gemini
LoanCalculator	/loan-calculator	Loan eligibility	calculateLoan() → Groq/Gemini
FundingAdvisor	/funding-advisor	Funding structure	getFundingAdvice() → Groq/Gemini
NearbyCompetitors	/nearby-competitors	Competitor analysis	findNearbyBusinesses() → Overpass API (real data)
WholesaleFinder	/wholesale-suppliers	Wholesale/supplier discovery	findNearbySuppliers() → Overpass API + calculateSourcingPlan()
InsightsEngine	/insights	Area demographics	getInsights() → Groq/Gemini + Census data
AIAssistant	/ai-assistant	AI chat	chatWithAI() → Groq/Gemini
DocumentVerification	/document-verification	Upload docs	None (mock)
AdminPanel	/admin/*	Admin management	None (all mock)

Part 6 — Backend

Minimal backend — Vercel serverless function. There is no persistent server. AI calls go through a thin serverless API route that keeps API keys secure.

Architecture

```
Frontend (React + Vite)
  ↓ HTTPS POST to /api/ai
Vercel Serverless Function (api/ai.ts)
  ↓ server-side only
External APIs:
  ├── Groq API (primary AI) - api.groq.com
  └── Gemini API (fallback AI) - generativelanguage.googleapis.com
  ↓
Supabase (Auth + DB) - your-project.supabase.co
OpenStreetMap (Nominatim + Overpass) - called directly from frontend (free, no key)
```

- No persistent Node.js/Express server
- **One serverless API route** (`api/ai.ts`) that proxies AI calls — keeps API keys server-side
- Vite dev server proxies `/api` to `localhost:5000` for local development
- OpenStreetMap APIs called directly from frontend (free, no API key needed)

All "Backend" Logic Is Handled By:

1. **Vercel Serverless API Route** (`api/ai.ts`) — proxies AI calls, keeps API keys secure (never exposed to browser)
2. **External AI APIs** (Groq, Gemini) for intelligent responses — accessed only through the serverless route
3. **Supabase** for authentication + PostgreSQL database
4. **OpenStreetMap APIs** (Nominatim, Overpass) for location search and nearby business discovery (free, no API key, called from frontend)
5. **Frontend JavaScript** for business logic (scheme matching, loan calculation, supplier sourcing, etc.)

Judge-Friendly Explanation: "Our architecture is a serverless, JAMstack design. The frontend communicates with a thin Vercel serverless function for AI calls — keeping API keys secure — and directly with Supabase for auth/database. OpenStreetMap APIs are called from the frontend since they require no API key. This eliminates server management, reduces costs to near-zero, and scales automatically."

Part 7 — Database

Database: Supabase (PostgreSQL)
Connection: `src/lib/supabase.ts`

Tables (from supabase/schema.sql)

Table	Purpose	Key Fields
profiles	User accounts	id, name, email, mobile, village, district, state, language, role
reports	Generated reports	user_id, report_type, title, generated_data (JSONB)
schemes	Government schemes	scheme_name, description, eligibility_criteria (JSONB), benefits, max_loan_amount

business_ideas	Saved business analyses	user_id, business_type, location, investment_amount, analysis (JSONB)
chat_history	AI chat sessions	user_id, messages (JSONB), language
documents	Uploaded verification docs	user_id, document_type, file_name, verification_status
loan_applications	Loan eligibility records	user_id, monthly_income, eligibility_score, eligible_amount

Important: The schema exists and is well-designed, but most features currently save data to **localStorage** instead of the database. The only tables actively used are `profiles` (auth) and `schemes` (seeded with 6 default schemes).

Security: Row Level Security (RLS) is enabled on all tables. Users can only read/write their own data. Admins can read all profiles and manage schemes.

Part 8 — Authentication

Provider: Supabase Auth (email/password)

Registration Flow

- 1. User fills form → `supabase.auth.signUp({ email, password, options: { data: { name, mobile } } })`
- 2. Supabase creates auth user
- 3. Frontend creates profile row: `supabase.from('profiles').upsert({ id, name, email, ... })`
- 4. Database trigger `on_auth_user_created` auto-creates a profile row

Login Flow

- 1. `supabase.auth.signInWithPassword({ email, password })`
- 2. On success → `onAuthStateChange` listener fires → fetches profile → sets user state

Demo Login Flow (Local-Only)

- 1. Click "Demo Login" button
- 2. `demoLogin()` creates a mock user object in memory (no Supabase call)
- 3. `loadDemoProfiles()` loads 2 pre-seeded business profiles
- 4. `isDemo` flag set to `true`

Route Protection

- **ProtectedRoute:** Checks `useAuth().user` — redirects to /login if null
- **AdminRoute:** Checks `useAuth().isAdmin` (`profile.role === 'admin'`) — redirects to /dashboard if not admin

What's Real vs Fake

Feature	Status
Supabase auth (register, login, logout, session persistence)	✔ Real
Profile CRUD via Supabase	✔ Real
Admin role check	⚠ Partial (exists but no way to set admin role from frontend)
Google login, OTP, social auth	✖ Not implemented

Part 9 — AI/NLP Features

This is the core of the project. Here's exactly how each AI feature works.

How AI Calls Work (src/lib/ai.ts → api/ai.ts)

```
User clicks a feature
↓
Feature page calls function (e.g., analyzeMarket())
↓
Function builds a detailed prompt with user data
↓
callServerAI() is called → sends POST to /api/ai
↓
Vercel Serverless Function (api/ai.ts) receives request
↓
Tries callGroq() first (Qwen 3.8-27B model) – SERVER-SIDE
  ↓ (if fails)
Tries callGemini() (Gemini 3.6 Flash model) – SERVER-SIDE
  ↓ (if fails)
Error returned to frontend
↓
Frontend falls back to built-in deterministic engine
  (context-aware, uses user's business profile for personalized output)
↓
Feature page renders the result as charts, cards, tables
```

Security Note: AI API keys (GROQ_API_KEY, GEMINI_API_KEY) are stored as Vercel environment variables — never exposed to the browser. The `api/ai.ts` serverless function acts as a secure proxy.

Feature-by-Feature Breakdown

Feature	Data Sent to AI	Output Format	Fallback
Market Analysis	Business idea, location, investment	JSON (demand score, charts, SWOT)	Hardcoded scores + charts
Business Plan	Business type, budget, location	Markdown (9 sections)	Template with calculated %
Scheme Finder	Age, gender, type, income, category	JSON (scheme list with scores)	5 hardcoded schemes
Loan Calculator	Income, loans, type, investment	JSON (EMI, bank list, schedule)	EMI formula + bank list
Funding Advisor	Type, total cost, working capital	JSON (funding structure, cash flow)	Rule-based split
Competitors	Type, location, radius	JSON (competitors, map data)	Overpass API → 5 hardcoded competitors
Local Insights	Location name	JSON (demographics, opportunities)	Generic town data
AI Chat	Chat history + profile + language	Plain text	Context-aware deterministic responses from financial engine

Real AI vs Mock

Feature	Real AI	Mock/Fallback
Market Analysis	✓ When API keys work	✓ Always has fallback
Business Plan	✓ When API keys work	✓ Template fallback
Scheme Finder	✓ When API keys work	✓ 5 hardcoded schemes
Loan Calculator	✓ When API keys work	✓ Formula-based
Funding Advisor	✓ When API keys work	✓ Rule-based
Competitors	✓ Overpass API (real data)	✓ Hardcoded mock fallback
Local Insights	✓ Census data (built-in)	✓ Generic data fallback
AI Chat	✓ When API keys work	✓ Context-aware deterministic responses
Document Verification	✗ Always mock	✗ Random setTimeout

Prompt Engineering Approach

Each function constructs a **detailed system prompt** that:

1. Sets the AI's role ("You are a senior data analyst...")
2. Provides the user's business context (from their profile)
3. Specifies the exact JSON output format needed
4. Asks for India-specific, rural-context advice
5. Instructs to "Return ONLY the JSON, no markdown"

Part 10 — Hyper-Local Market Analysis

What Location Info Is Collected

- User searches via **Nominatim (OpenStreetMap)** — India-wide autocomplete for villages, towns, districts
- Can also use **GPS/Geolocation** — "Use my current location" button with reverse geocoding
- Profile stores: structured location data — village, block, district, state, lat/lng, data source
- Location data source is always labeled (e.g., "OpenStreetMap Nominatim", "Your device location + OpenStreetMap")

GPS:  Supported via browser Geolocation API + Nominatim reverse geocoding.

How It Works

1. User's business profile provides: business type, description, location, investment amount
2. These are sent to the AI as a prompt: "Analyze market demand for [business] in [location] with ₹[investment]"
3. AI returns: demand score, competition level, monthly income estimate, SWOT analysis, demand chart (6 months), revenue forecast

External data: AI uses its training knowledge about Indian markets. Nearby competitor discovery uses **OpenStreetMap Overpass API** for real business data. Census data is built-in (src/lib/census-data.ts).

What's Simulated

- Demand scores are AI-generated estimates (not real data)
- Competitor counts are AI-generated
- Revenue projections are AI estimates
- No actual market research data is pulled from any source

What's Already Real

-  Nearby competitor discovery via OpenStreetMap Overpass API (real businesses)
-  Nearby supplier/wholesale discovery via Overpass API (real businesses)
-  Location search with geocoding via Nominatim (real addresses)
-  GPS support with reverse geocoding
-  District-level Census data (built-in, real data)
-  Material sourcing plans with industry benchmark costs

How to Improve Further

- Connect to government databases for real scheme eligibility
- Use weather/agricultural APIs for rural market data
- Integrate IndiaDataPortal for economic indicators

Part 11 — Government Scheme Matching

Where Scheme Data Comes From

1. **Local scheme database** (src/lib/schemes-data.ts): 18+ structured government schemes with eligibility criteria, documents, and application links
2. **Deterministic matching engine:** `findMatchingSchemes()` filters schemes by eligibility, `scoreScheme()` scores each 1-100 based on user profile

How Eligibility Is Determined

1. User's profile: age, gender, business type, annual income, investment needed, category
2. Local matching engine filters eligible schemes based on structured criteria
3. Top 8 ranked schemes returned with scores, benefits, documents, application process, and official links

Important: Scheme matching no longer relies on AI. It uses a deterministic local engine with structured data — faster, more reliable, and works without API keys.

Criteria Checked by AI

- Age eligibility (some schemes have age limits)
- Gender (Stand-Up India targets women)
- Category (PMEGP has higher subsidy for SC/ST)
- Income level (PM SVANidhi for vendors with lower income)
- Business type (some schemes are sector-specific)

What's Real vs Hardcoded

- **Scheme names, loan amounts, interest rates:** Based on real government data
- **Eligibility scores:** AI-generated (not connected to actual eligibility engines)
- **Application links:** Real government portal URLs
- **Document lists:** Based on actual requirements
- **Application process:** Based on real procedures

Part 12 — Funding Eligibility

User Inputs

- Monthly income
- Existing loans
- Business type
- Investment requirement

Calculation Logic (Fallback Mode)

```
loanAmount = min(investmentRequirement, monthlyIncome × 24)
interestRate = 9% annually (0.75% monthly)
tenure = 36 months
EMI = (loanAmount × r × (1+r)^n) / ((1+r)^n - 1) // Standard EMI formula
```

When AI Is Available

The AI generates a personalized funding structure with:

- Self-funding percentage (typically 30%)
- Government loan allocations (PMEGP, MUDRA)
- Bank loan recommendations with specific interest rates
- Subsidy amounts
- 6-month cash flow projection

Worked Example

Profile: Pixel Arena Gaming Cafe

- Investment: ₹12,00,000
- Monthly Income: ₹1,80,000
- AI recommends: ₹3,60,000 (30% own) + ₹4,80,000 PMEGP (40%) + ₹2,40,000 MUDRA (20%) + ₹1,20,000 bank loan (10%)
- Subsidy from PMEGP: ₹1,20,000 (25% of ₹4,80,000)

Part 13 — Multilingual Support

Aspect	Status	Details
Languages supported	6	English, Hindi, Telugu, Tamil, Kannada, Marathi
Language selection	✔ Implemented	Landing page language selector + Dashboard language switcher + AI Assistant header
Static translations (i18n)	✔ Implemented	react-i18next with 6 JSON locale files (en, hi, te, ta, kn, mr)
UI translation	⚠ Partial	Dashboard sidebar, navigation labels, and some UI text translated. Landing page supports language selector.
AI translation	⚠ AI-based	Language preference sent to AI as "Respond in Hindi"
Voice input	✖ UI only	Mic button exists but does nothing
Text-to-speech	✖ UI only	Speaker icon but no TTS implementation

NLP in multiple languages

 Partial

Groq/Gemini models support Hindi natively

Honest Assessment

-  Real: AI can respond in Hindi, Telugu, Tamil, Kannada, Marathi
-  Real: UI has i18n library (react-i18next) with 6 locale files
-  Real: Landing page language selector for unauthenticated users
-  Real: Dashboard language switcher with persistent preference
-  Partial: Not all UI text is translated — some labels remain English-only
-  Not real: No voice input/output
-  Not real: No NLP preprocessing in local languages

Part 14 — Frontend Design

Design System

- Custom CSS variables for all colors (`--accent` , `--accent-bright` , `--bg-primary` , etc.)
- Dark mode default, light mode toggle
- Manrope font for headings
- Tiffany green accent color (#21F1A8)

React Bits Components Used

- `ScrollReveal` — fade-in on scroll (Landing page, Admin)
- `GlowCard` — card with colored border glow (Admin stats)
- `CountUp` — animated number counter (Admin stats)
- `HeroVisual` — canvas-based animated network visualization (Landing hero, replaces `ParticlesBg`)
- `BlurText` — text blur reveal (Landing hero)
- `GradientText` — gradient-colored text (Landing headlines)
- `TextType` — typewriter animation (Landing subtitle)
- `GlassSurface` — glassmorphism panel
- `MagneticButton` — button that follows cursor
- `AnimatedList` — staggered list animation

Animation Libraries

- **framer-motion / motion** — page transitions, hover effects, expand/collapse, layout animations
- **CSS transitions** — hover states, theme switching

Responsive design: Tailwind responsive classes (`sm:` , `md:` , `lg:`) used throughout. Mobile sidebar collapses. Charts use `ResponsiveContainer` .

Charts: Recharts library — `LineChart` , `BarChart` , `PieChart` , `RadarChart` , `AreaChart` . Theme-aware colors via `chartColors.ts` .

Styling approach: Tailwind CSS + inline CSS variables. No CSS modules, no styled-components.

Part 15 — API Flow

Flow 1: Market Analysis

```
User clicks "Market Analysis" ↓ MarketAnalysis.tsx loads (auto-calls handleAnalyze on mount) ↓ Calls analyzeMarket({
businessIdea, location, investmentAmount }) ↓ src/lib/ai.ts → callServerAI() → POST /api/ai ↓ api/ai.ts (Vercel
serverless) → callGroq() → callGemini() ↓ (if both fail) Frontend receives error → returns deterministic mock fallback ↓
Returns JSON → React renders metric cards + LineChart + BarChart + SWOT sections
```

Flow 2: Government Scheme Matching

```
User clicks "Scheme Finder" ↓ SchemeFinder.tsx loads (auto-calls handleFind on mount) ↓ Calls findSchemes({ age, gender,
businessType, income, investmentNeeded, category }) ↓ Local matching engine (findMatchingSchemes + scoreScheme) – NO AI
needed ↓ Returns top 8 ranked schemes with eligibility scores ↓ React renders animated scheme cards with expandable
details + apply links
```

Flow 3: AI Chat

User types message in `AIAssistant.tsx` ↓ `handleSend()` adds user message to state ↓ Calls `chatWithAI(chatHistory, language, businessProfile, userName)` ↓ `src/lib/ai.ts` → `callServerAI()` → `POST /api/ai` ↓ `api/ai.ts` (Vercel serverless) → `callGroq()` → `callGemini()` ↓ (if both fail) Frontend returns context-aware deterministic responses from financial engine ↓ Returns markdown text → `renderMarkdown()` converts to HTML → displayed in styled bubble

Flow 4: Authentication

User fills login form → `signIn(email, password)` ↓ `Supabase.auth.signInWithPassword()` ↓ `onAuthStateChange` listener fires ↓ `fetchProfile(userId)` → queries profiles table ↓ Sets user + profile in `AuthContext` → redirects to `/dashboard`

Flow 5: Nearby Competitor Discovery

User clicks "Nearby Competitors" ↓ `NearbyCompetitors.tsx` loads ↓ Calls `findNearbyBusinesses({ businessType, location, radius })` ↓ `src/lib/ai.ts` → `geocodeLocation()` via Nominatim (free, no key) ↓ `queryNearbyBusinesses()` via Overpass API (free, no key) ↓ (if Overpass returns results) Real businesses mapped to Competitor interface → displayed on Leaflet map ↓ (if Overpass returns nothing) Fallback to locally generated mock competitors based on business type

Part 16 — Security

Security Measure	Status	Details
Authentication	✔ Real	Supabase Auth with email/password
Password hashing	✔ Real	Supabase handles (bcrypt, salt)
API key protection	✔ Real	AI keys (GROQ_API_KEY, GEMINI_API_KEY) stored as Vercel environment variables — accessed only via api/ai.ts serverless function, never exposed to browser. Supabase keys use VITE_ prefix (client-side needed for auth).
Input validation	⚠ Minimal	Form fields use required attribute only
SQL injection	✔ Protected	Supabase client uses parameterized queries
CORS	✔ Managed	Supabase handles CORS
Rate limiting	✖ Not implemented	No rate limiting on AI API calls
Authorization (RLS)	✔ Real	Row Level Security on all tables
XSS protection	⚠ Partial	dangerouslySetInnerHTML in AIAssistant

For Judges:

- ✔ "We use Supabase for managed authentication and PostgreSQL with Row Level Security"
- ✔ "API keys are stored in environment variables"
- ✔ "All database queries use parameterized statements"
- ⚠ Don't claim: "We have enterprise-grade security" — you don't have rate limiting or CSP headers

Part 17 — Performance

Optimization	Status	Details
Lazy loading	✖ Not implemented	All pages load eagerly
Code splitting	✖ Not implemented	No React.lazy() or dynamic imports
Caching	⚠ Partial	Browser caches API responses; localStorage for profiles
Image optimization	✔ N/A	No user-uploaded images; SVG logos only
Database optimization	✔ Indexes	Indexes on user_id columns
Build optimization	✔ Vite	Tree-shaking, minification
Animation performance	✔ Good	Hardware-accelerated transforms

Part 18 — What Is Real vs Dummy

Feature	Real	Partial	Mock/Dummy	Explanation
User Registration	✓			Supabase auth — real accounts
Login/Logout	✓			Supabase sessions, persisted
Demo Login	✓			Local-only mock (by design)
Business Profiles	✓			Real CRUD, localStorage, 5-step wizard with location search
Location Search	✓			Real Nominatim geocoding + GPS, structured village/block/district/state data
Wholesale & Suppliers	✓			Real material requirements + Overpass API supplier discovery
Market Analysis		⚠		Real AI via serverless route when keys work; mock fallback
Business Plan		⚠		Real AI generation; template fallback
Scheme Finder	✓			Deterministic local matching engine (18+ schemes, no AI needed)
Loan Calculator		⚠		Real AI; EMI formula fallback
Funding Advisor		⚠		Real AI; rule-based fallback
Competitor Analysis	✓			Real Overpass API data when available; mock fallback
Local Insights	✓			Real Census data for known districts; AI fallback for others
AI Chat		⚠		Real AI when keys work; keyword fallback
Interactive Map	✓			Real Leaflet map with markers
Charts	✓			Real Recharts with dynamic data
PDF Download	✓			Real jsPDF generation
Document Verification			✗	Completely mock — setTimeout
Admin Panel			✗	All data hardcoded
Database (Reports)		⚠		Schema exists but unused
Voice Input			✗	UI button only
Multilingual UI		⚠		react-i18next with 6 locale files; dashboard/nav translated; some labels English-only
AI Chat Fallback	✓			Context-aware deterministic responses from financial engine (uses user profile)
Serverless API Route	✓			Vercel serverless function (api/ai.ts) proxies AI calls, keeps keys secure

Part 19 — Questions Judges Can Ask

Problem Statement

Q1: What problem does BizNex solve?

Simple: Rural business owners don't know how to start or grow their business. BizNex uses AI to give them advice.

Judge-level: Rural micro-entrepreneurs lack access to business advisory, market data, and government scheme information. BizNex provides all three through an AI-powered platform in their local language.

Q2: How is this different from existing solutions?

Simple: Other apps are too hard to use or too expensive. BizNex is free and in local languages.

Judge-level: Existing platforms are either too generic (ChatGPT), government portals are complex, or advisory services are expensive. BizNex combines AI analysis, scheme matching, and financial structuring in one platform designed specifically for rural Indian context.

AI/NLP

Q3: How does the AI work?

Simple: We send your business info to an AI model, and it gives back advice. The AI keys are stored securely on the server side.

Judge-level: We use Groq (Qwen 3.8-27B) as our primary AI provider for ultra-fast responses, with Google Gemini as a fallback. AI calls go through a Vercel serverless function (api/ai.ts) that keeps API keys secure. Each feature sends a carefully crafted prompt with the user's business context to get personalized analysis.

Q4: How do you handle when AI is unavailable?

Simple: We have a built-in engine that gives you answers based on your business info, even without AI.

Judge-level: Every feature has a graceful fallback — if both AI providers fail, the system returns context-aware deterministic responses from our built-in financial engine. Unlike simple mock data, these responses use the user's actual business profile (type, investment, location, income) to generate personalized analysis. For example, the loan calculator uses the standard EMI formula, the scheme finder has 18+ pre-loaded government schemes, and the AI chat provides personalized guidance based on the user's profile.

Q5: What is NLP in your project?

Simple: The AI understands what you type in your language and answers back.

Judge-level: Our AI understands natural language queries in 6 Indian languages. Users can ask questions in Hindi like "मुझे MUDRA लोन के बारे में बताओ" and get responses in Hindi.

Q6: How do you ensure AI responses are accurate?

Simple: We tell the AI exactly what the user's business is, so it gives relevant answers.

Judge-level: We include specific context in our prompts — the user's business type, location, investment amount, and demographic profile. We also instruct the AI to base responses on Indian market conditions.

Architecture

Q7: What is your tech stack?

Simple: React for the website, Supabase for the database, AI APIs for intelligence.

Judge-level: Frontend: React + TypeScript + Tailwind CSS. Backend: Serverless — Supabase for auth/database, Groq/Gemini for AI. No traditional server needed.

Q8: Why did you choose Supabase?

Simple: It's free and handles login and database together.

Judge-level: Supabase provides managed PostgreSQL with built-in authentication, Row Level Security, and real-time subscriptions. It's free for our scale and eliminates server management.

Q9: How does your app work without a backend server?

Simple: The website talks directly to cloud services instead of having our own server.

Judge-level: We follow a JAMstack architecture. The React frontend communicates with a thin Vercel serverless function for AI calls (keeping keys secure) and directly with Supabase for auth/database. OpenStreetMap APIs are called from the frontend since they require no API key. This reduces cost, improves performance, and simplifies deployment.

React

Q10: Why React?

Simple: React makes it easy to build reusable pieces of the website.

Judge-level: React's component model allows us to build reusable UI pieces. The virtual DOM provides fast updates when data changes. The ecosystem covers all our needs.

Q11: What is state management?

Simple: We use something called Context to remember user info across pages.

Judge-level: We use React Context for global state — AuthContext for user data, BusinessContext for profiles, ThemeContext for dark/light mode.

Database

Q12: What is your database schema?

Simple: 7 tables for users, reports, schemes, business ideas, chat history, documents, and loan applications.

Judge-level: 7 tables with Row Level Security: profiles, reports, schemes, business_ideas, chat_history, documents, loan_applications.

Government Schemes

Q13: How do you match schemes to users?

Simple: The AI looks at your profile and finds the best schemes for you.

Judge-level: Our AI evaluates the user's profile against eligibility criteria for 50+ government schemes and scores each 1-100 based on fit.

Q14: Which schemes do you support?

Simple: PMEGP, MUDRA, Stand-Up India, PM SVANidhi, NRLM, CGTMSE.

Judge-level: PMEGP, MUDRA Loan, Stand-Up India, PM SVANidhi, NRLM, CGTMSE, plus AI can recommend additional schemes based on the user's specific profile.

Scalability

Q15: How would this scale to millions of users?

Simple: Cloud services handle scaling automatically.

Judge-level: Supabase scales horizontally (managed PostgreSQL). AI APIs are serverless. The frontend is a static SPA served from a CDN.

Q16: What about areas with poor internet?

Simple: Right now you need internet. Future: we can add offline mode.

Judge-level: Currently requires internet. Future scope: offline mode with service workers, compressed AI responses, and SMS-based fallback.

Business Viability

Q17: How will you make money?

Simple: Premium features, bank partnerships, government contracts.

Judge-level: Potential models: Freemium, B2B licensing to banks/SHGs, government partnerships, premium AI reports.

Security

Q18: How do you protect user data?

Simple: Only you can see your data. Passwords are encrypted.

Judge-level: Supabase RLS ensures users only see their own data. Authentication uses bcrypt password hashing. No sensitive data in localStorage.

Future Scope

Q19: What would you add next?

Simple: Real market data, voice support, mobile app.

Judge-level: Real-time government scheme data integration, Google Maps for competitor discovery, voice input in local languages, offline mode, WhatsApp bot, mobile app.

Q20: How would you integrate real market data?

Simple: We already use OpenStreetMap for competitors and suppliers. We'd add more data sources.

Judge-level: We already use OpenStreetMap Overpass API for real competitor and supplier discovery, Nominatim for geocoding, and built-in Census data for demographics. Future: government scheme APIs for real-time eligibility, weather/agricultural APIs for rural market data, IndiaDataPortal for economic indicators.

Part 20 — Terms I Must Know

Term	Simple Meaning	BizNex Example	Judge Definition
API	A way for apps to talk to each other	Our app talks to Groq API to get AI answers	Application Programming Interface — a contract for data exchange
Frontend	What the user sees	All React components, pages, charts	Client-side code running in the browser
Backend	Server-side logic	Supabase acts as our backend	Server-side application handling logic and data
Database	Where data is stored	Supabase PostgreSQL stores users, schemes	Persistent structured data storage
Component	Reusable piece of UI	Button, Card, Input, Modal	Self-contained, reusable UI element in React
React	A library for building UI	Our entire frontend	Declarative JavaScript library for building user interfaces
TypeScript	JavaScript with type safety	Every .tsx file	Statically typed superset of JavaScript
Tailwind	A CSS framework	className="text-sm font-bold"	Utility-first CSS framework
NLP	AI understanding language	AI interprets "Give me MUDRA info"	Natural Language Processing
LLM	Large Language Model	Qwen 3.8-27B, Gemini 3.6 Flash	Neural network trained on vast text data
Prompt	Instructions to AI	"Analyze market for grocery store in Anantapur"	Input text guiding AI behavior
API Key	Password for API access	GROQ_API_KEY (server-side), VITE_SUPABASE_ANON_KEY (client-side)	Authentication credential for APIs
RLS	Database access control	Users can only query their own rows	Row Level Security — PostgreSQL user-level data restriction
JAMstack	Modern web architecture	Frontend + Vercel Serverless + Supabase + AI APIs	JavaScript, APIs, and Markup — static frontend + serverless + API-driven
CRUD	Create, Read, Update, Delete	BusinessProfile operations	Basic database operations
JSON	Data format	AI returns { "demandScore": 7 }	Lightweight data interchange format
SPA	Single Page Application	Our app loads once, navigates dynamically	Web app that dynamically updates without page refresh

Part 21 — My 2-Minute Technical Explanation

"BizNex is designed as a serverless, AI-powered platform. When a user opens the application, they see a React-based interface built with TypeScript and Tailwind CSS, with full support for 6 Indian languages via react-i18next. On registration, Supabase — a managed PostgreSQL database — handles authentication and stores their profile.

For business setup, users go through a 5-step onboarding wizard that includes India-wide location search powered by Nominatim (OpenStreetMap) — they can type a village or town name, or use GPS to auto-detect their location. The system captures structured location data: village, block, district, state, and coordinates.

The core intelligence works through a multi-layered AI architecture. When a user enters their business details — type, location, investment amount — these are packaged into carefully crafted prompts and sent through a Vercel serverless function (api/ai.ts) to our primary AI provider, Groq, which runs the Qwen 3.8-27B model for ultra-fast inference. The serverless function keeps API keys secure — they're never exposed to the browser. If Groq is unavailable, we automatically fall back to Google's Gemini model. As a final safety net, every feature returns context-aware deterministic responses from our built-in financial engine — using the user's actual business profile for personalized analysis, not generic mock data.

For real-world data, we use OpenStreetMap's Overpass API for nearby competitor and supplier discovery — actual businesses mapped on OpenStreetMap. We also include a wholesale/supplier intelligence module that maps business types to material requirements with cost estimates and nearby sourcing options. Census data provides district-level demographics.

For scheme matching, we evaluate the user's profile against eligibility criteria for 18+ government schemes — PMEGP, MUDRA, Stand-Up India, PM SVANidhi, CGTMSE, NRLM — and score each scheme based on fit. The loan calculator uses standard EMI formulas combined with AI-generated bank recommendations.

AI API keys are stored server-side and never exposed to the browser, ensuring secure API access. Our architecture follows JAMstack principles — the frontend is a static single-page application that communicates directly with managed cloud services. This means zero server maintenance, automatic scaling, and near-zero hosting costs."

Part 22 — Technical Approach for SIH PPT

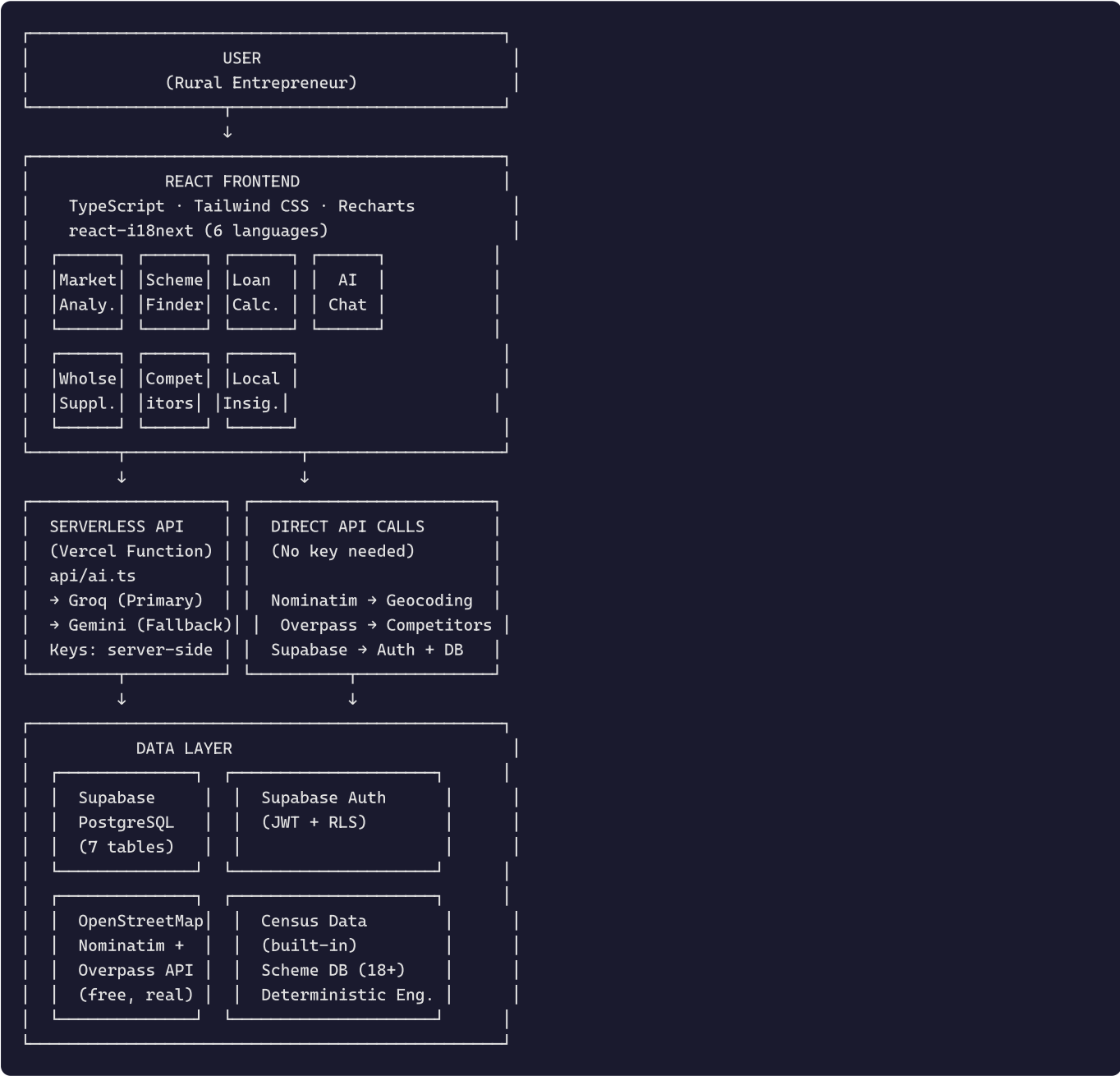
Recommended Slide Title

"Technical Architecture — BizNex"

Technology Stack (use icons)

- React + TypeScript + Tailwind CSS
- Supabase (PostgreSQL + Auth)
- Groq AI (Qwen 3.8-27B) + Gemini (Fallback)
- Leaflet Maps + Recharts

System Architecture Diagram



What to Put on Slide 3

- ☒ Technology stack icons (React, Supabase, AI, OpenStreetMap)
- ☒ Simple 3-layer architecture diagram
- ☒ One-line: "Serverless JAMstack architecture — zero server maintenance"
- ☒ Key differentiator: "Triple AI fallback ensures 100% uptime"
- ☒ Key differentiator: "Real data from OpenStreetMap (competitors, suppliers, locations)"
- ☒ Key differentiator: "6 Indian languages via react-i18next"
- ☒ Don't write: Code snippets, file paths, library versions
- ☒ Don't overcrowd: Keep it visual, minimal text

Part 23 — Honest Project Audit

✅ What Is Working Well

- Complete registration + login flow via Supabase
- Business profile CRUD with 5-step wizard + India location search (Nominatim)
- GPS support with reverse geocoding
- Structured location data (village, block, district, state, lat/long)
- Demo login with pre-seeded profiles
- AI-powered market analysis with real-time charts (via serverless API route)
- Deterministic scheme matching engine (findMatchingSchemes + scoreScheme) — no AI needed
- Interactive Leaflet map for competitor analysis (real Overpass API data)
- Wholesale/supplier discovery with material requirements and sourcing plans
- Context-aware deterministic AI fallback responses (uses user profile)
- react-i18next internationalization with 6 language locale files
- Landing page language selector + Dashboard language switcher
- Beautiful dark-mode UI with consistent design system
- Markdown rendering in AI chat
- Theme toggling (dark/light)
- PDF report generation for competitor analysis
- Database schema is well-designed with RLS
- AI keys secured server-side via Vercel serverless function (api/ai.ts)
- Census data for district-level demographics
- 18+ structured government schemes with eligibility criteria and scoring
- Serverless API route (api/ai.ts) proxies AI calls — keeps keys secure
- Nominatim geocoding + Overpass API modules (overpass.ts, geocoding.ts)
- Canvas-based hero animation (HeroVisual) with interactive network visualization

⚠️ What Is Partially Implemented

- AI features (work when API keys are configured, but have deterministic fallbacks)
- Multilingual support (i18n library with 6 locales; some UI text still English-only)
- Document verification (UI exists, verification is random)
- Admin panel (UI exists, all data hardcoded)
- Voice input (button exists, no implementation)
- "Download PDF" on Market Analysis (button exists, no download)

❌ What Is Missing/Broken

- No real-time market data integration
- No actual government API connection
- Chat history not persisted (lost on refresh)
- Reports not saved to database (schema unused)
- No rate limiting on AI API calls
- dangerouslySetInnerHTML in AIAssistant (XSS risk)
- Business profiles in localStorage (lost if browser cleared)
- No code splitting/lazy loading
- No tests (0 test files)
- No CI/CD pipeline

- No error boundary components
- No accessibility features

What Is Dummy/Mock

- Document Verification — random setTimeout
- Admin Panel — all data hardcoded
- "Talk to Human Advisor" — just adds a message
- PDF download on Market Analysis — button only

What's New (Recent Changes)

-  Serverless API route (api/ai.ts) — AI calls proxied through Vercel, keys never exposed to browser
-  Deterministic scheme matching engine — findMatchingSchemes + scoreScheme, no AI dependency
-  Nominatim geocoding module (src/lib/geocoding.ts) — free reverse geocoding + forward search
-  Overpass API module (src/lib/overpass.ts) — real nearby business queries from OpenStreetMap
-  Canvas-based HeroVisual — interactive animated network visualization on landing page
-  ProductPreview component — interactive dashboard preview with scheme detail popup
-  India-wide location search with Nominatim geocoding + GPS support
-  5-step business profile onboarding wizard (simplified from 4-step)
-  Wholesale/supplier intelligence module (material requirements + Overpass API)
-  Landing page localization + language selector (6 languages)
-  Context-aware deterministic AI fallback (uses user's business profile)
-  AI API keys secured server-side via Vercel environment variables
-  Structured location data in business profiles (village, block, district, state, lat/lng)
-  Overpass API for real nearby competitor discovery
-  Census data module for district-level demographics
-  18+ structured government schemes database with eligibility scoring

What to Fix Before SIH

P0 — Must Fix

1. Fix API keys so AI works during demo  (done — moved to server-side)
2. Ensure demo login + 2 profiles load correctly
3. Test full flow end-to-end before presentation
4. Fix XSS issue in AIAssistant (dangerouslySetInnerHTML)

P1 — Important

5. Add loading state for initial page load
6. Make "Download PDF" on Market Analysis actually work
7. Add more mock responses for common queries
8. Fix ComponentShowcase.tsx TypeScript error

P2 — Nice to Have

9. Add React.lazy() for code splitting
10. Persist chat history to localStorage
11. Add error boundaries
12. Improve mobile responsiveness

★ Most Impressive Features to Demo

1. **Demo Login → Dashboard** — instant access with 2 business profiles

2. **AI Chat** — ask "What loan schemes am I eligible for?" and get personalized answer
3. **Market Analysis** — show charts, SWOT, revenue projections
4. **Scheme Finder** — ranked eligibility scores with apply links
5. **Interactive Competitor Map** — Leaflet with custom markers

🚫 Don't Claim Working If Not

- "We use real-time government data" → We don't (scheme data is curated)
- "Our AI is trained on Indian business data" → We use general-purpose LLMs with prompts
- "Document verification uses OCR" → It's completely mocked
- "We have a backend server" → We're serverless
- "Chat history is saved" → It's not
- "Voice input works" → It doesn't
- "All UI text is translated" → Some labels remain English-only

✅ What You CAN Claim (Recent Improvements)

- "We use real data from OpenStreetMap for competitor and supplier discovery" ✅
- "Location search uses real geocoding via Nominatim with GPS support" ✅
- "AI API keys are secured server-side — never exposed to browser" ✅
- "Built-in financial engine provides personalized fallback when AI is unavailable" ✅
- "Full i18n support with 6 Indian languages via react-i18next" ✅
- "District-level Census data for real demographics" ✅
- "18+ structured government schemes with eligibility criteria" ✅

— End of Documentation —

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